

Technical Architecture Report — Proximie Live OR Action Segmentation

Senior ML Engineer Challenge • July 2026

Executive summary. Separate real-time clinical inference from retention and annotation. Treat raw video as an ephemeral streaming input; use compact visual embeddings, quality metadata, uncertainty and boundary signals downstream. Target the two known failure modes with quality-aware multi-view fusion, multi-scale temporal context, and explicit boundary refinement.

1. Data Strategy & Annotation Lifecycle

In-flight hard-example capture

Emit a privacy-reviewed derived event containing a pseudonymous session ID, time bucket, class probabilities, confidence margin, camera quality, inter-camera disagreement, boundary confidence and model version. Trigger events for low confidence, oscillation, disagreement, rare classes, implausible transitions and unstable Operation boundaries. Do not create a hidden debug-video copy. If an approved investigation needs raw media, use an authorized, encrypted, audited, short-TTL workflow. Embeddings still require governance because they can remain sensitive.

Active learning over historical terabytes

Run a cheap embedding pass and score temporal windows using uncertainty, instability, camera disagreement, transition likelihood, rarity and diversity. Select candidates with k-center/farthest-point sampling and quotas by hospital, camera configuration and workflow position. Annotate short temporal clips around uncertain boundaries rather than isolated frames.

Annotation optimization

Use self-supervised temporal pre-training, then a small supervised seed model, confidence-filtered pseudo-labels, and targeted human correction. Prioritize boundary annotations and optimize model improvement per annotation minute.

2. Modeling & Multi-Sensor Fusion

Multi-view spatial fusion

Calibrate fixed cameras with intrinsics/extrinsics and a shared OR coordinate system. Initially fuse per-camera embeddings using quality weights derived from visibility/blur/occlusion signals, while appending mean/min quality and camera disagreement. A later version can use cross-view attention with camera-position embeddings. Feature fusion is the practical first step because it is cheaper and more tolerant of surgical occlusion than pixel registration.

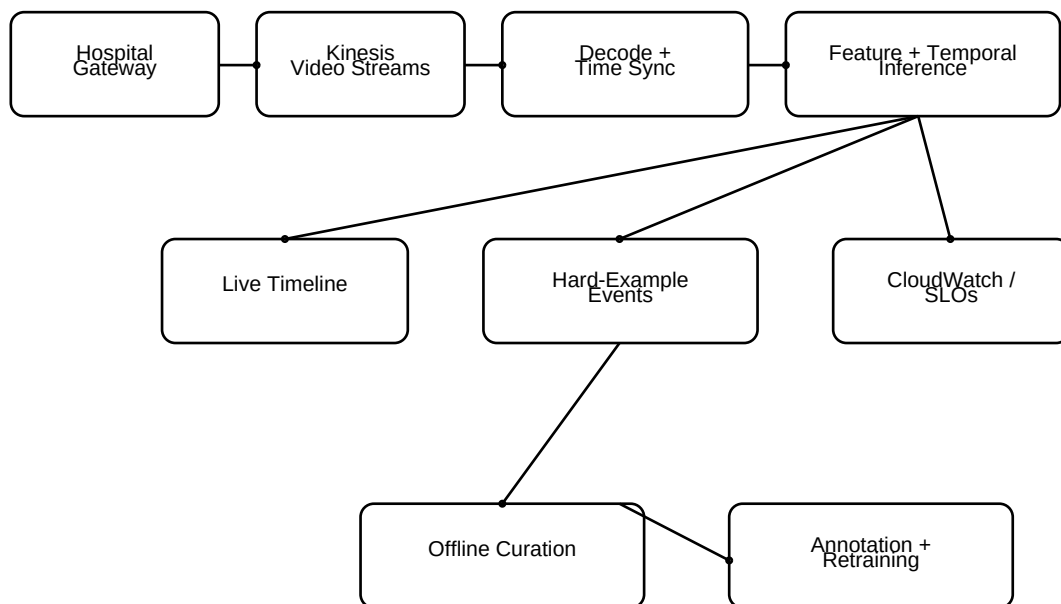
Temporal scale balancing

Use two temporal branches: a high-resolution stream for rapid Operation transitions and a downsampled long-context stream for Patient Present. Dilated temporal convolutions or efficient local attention give predictable compute. Add a dedicated start/end boundary head and class-specific duration constraints in the decoder/post-processor.

3. AWS Cloud Architecture & Cost Optimization

The real-time path uses managed video ingestion, bounded synchronization, shared inference workers, and a derived event stream. Raw-video retention is explicitly governed rather than treated as an implicit debugging mechanism.

Proposed AWS scale architecture



Concrete AWS mapping

Kinesis Video Streams for live video ingestion; **ECS on GPU EC2** or **EKS GPU nodes** for a shared inference fleet; **SageMaker real-time endpoints** as an alternative managed serving plane; **Kinesis Data Streams** for derived feature/event fan-out; **S3** for approved offline datasets/artifacts; **DynamoDB** for session/configuration state; **CloudWatch** for latency, GPU, dropped-frame and model-health metrics; **IAM/KMS** for least privilege/encryption; **EventBridge/Step Functions** for offline curation/retraining orchestration.

Synchronization and jitter

Every frame carries capture timestamp, gateway timestamp, camera ID and sequence number. A bounded per-camera jitter buffer aligns frames within a tolerance window. Late frames do not block indefinitely: mark the camera degraded, optionally reuse a recent embedding briefly, and expose the quality state to the model.

Cost optimization

Use a cheap always-on model and invoke expensive inference around uncertain windows. Reduce sampling during confidently static periods and increase it near boundaries. Pack multiple ORs per accelerator and micro-batch independent feature inference. Autoscale on queue depth/GPU utilization with cooldown hysteresis. Finally, distill and quantize the model after validating rare-class recall and boundary metrics.

4. Evaluation Strategy

Report frame accuracy, macro/per-class F1, segment IoU, boundary MAE in seconds, fragmentation rate, Patient Present false-positive seconds, missed-operation duration and transition recall within a tolerance window. Make Patient Present FP duration and Operation boundary MAE primary product metrics. Split evaluation by hospital, room, camera configuration and time period to prevent correlated-frame leakage.

5. Prototype Mapping & Trade-offs

The repository maps production interfaces into small modules: mock ingestion, quality-aware fusion, GRU temporal classification, deterministic segment cleanup, dual metrics and targeted error analysis. Feature-level fusion is cheaper than pixel fusion; the small GRU is intentionally reviewable; rule-based cleanup is transparent but should later be compared with a learned/semi-Markov decoder.

Recommended next experiments

Benchmark GRU vs dilated TCN vs local attention at equal latency; add/ablate a boundary head; compare camera fusion strategies under occlusion; simulate active-learning improvement per labeled minute; establish latency/synchronization/fragmentation SLOs; validate retention-safe hard-example handling with privacy/security stakeholders.